Beyond clever presentation: Volkswagen’s emissions test

Team 5 · When did a company seriously manipulate its data?

What did VW do to the test? What was it trying to achieve? Who paid the price?

The claim and the measurement

The claim

Compliance with U.S. standards

Vehicles marketed as clean diesel
Certification supported market access

The quantity

NOx mass per distance

Nitrogen oxides in the exhaust
Measured in grams per kilometre

Reference standard: 0.043 g/km NOx at full useful life

WVU/ICCT (2014), Table 2.2 p. 5 (0.07 g/mi), Table 4.1 p. 59 (0.043 g/km); VW plea (2017), Exh. 2, ¶¶ 28–30, 32 (Exh. 2 pp. 10–12).

Laboratory and road measurements

Laboratory

Prescribed conditions

A chassis dynamometer simulates driving
FTP-75 uses a standard speed sequence

Road

Changing conditions

PEMS measures exhaust on a moving car
Traffic, gradient and load vary

The standard is a reference for road results; road routes are not certification cycles.

WVU/ICCT (2014), pp. 9, 39–55, 59, 64.

Three vehicles, with limited coverage

Vehicle A

2.0 L

Lean-NOx trap

Vehicle B

2.0 L

Urea-SCR

Vehicle C

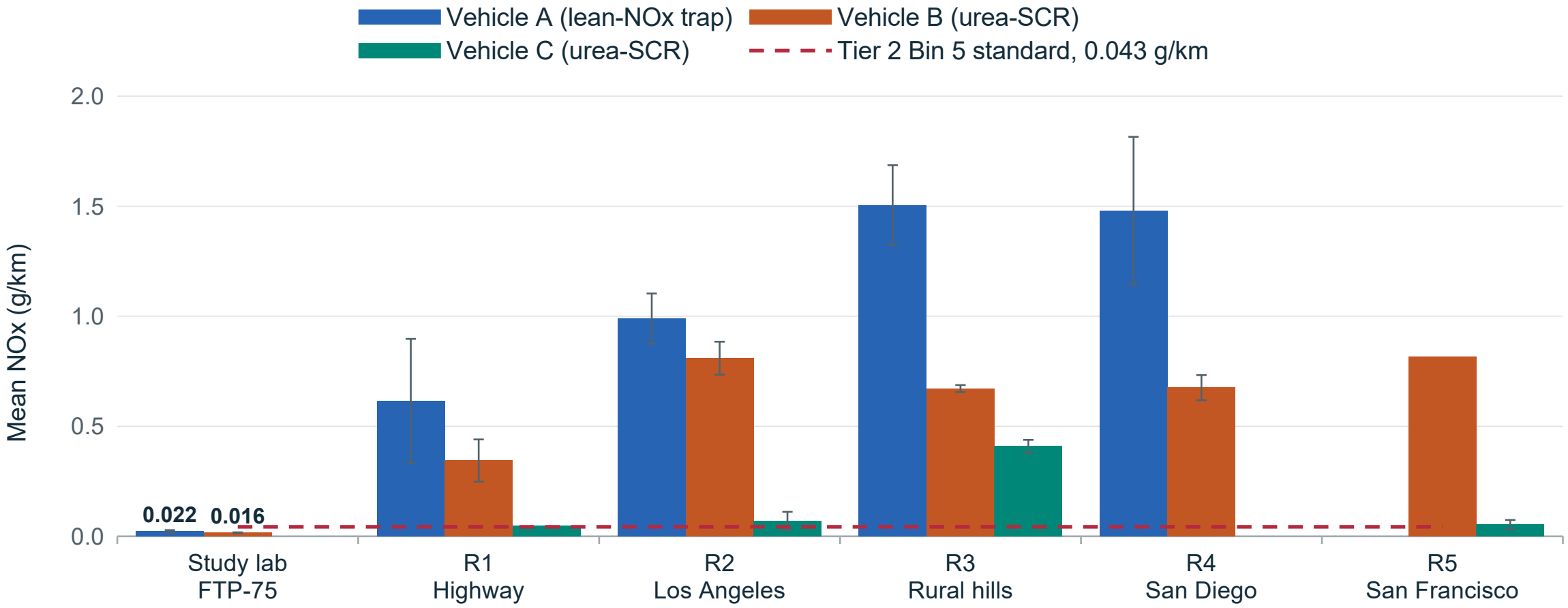
3.0 L

Urea-SCR

Road tests: A, B, C Laboratory tests: A, B
Three case studies do not estimate fleet-wide prevalence.

WVU/ICCT (2014), p. iv; pp. 1, 9–10; Table 3.15 p. 57. EPA NOV (2015), p. 4 (2012 Jetta, 2013 Passat).

The measured NOx results



Bars: route or test means; thin bars: ±1 SD as published (none for single runs). No bar = no test (C lab, C R4, A R5), not zero.

The standard is a certification benchmark, used here as a reference; road routes are not certification tests. Three vehicles: no fleet-wide estimate.

WVU/ICCT (2014), Table 4.3 p. 65; p. 64; Table 4.1 p. 59.

The denominator changes the meaning

35.0×

1.505 ÷ 0.043

Vehicle A, rural road mean relative to the standard

≈68×

1.505 ÷ 0.022

The same road mean relative to A's lab mean (0.022 ± 0.006, two tests)

Report summary: A ≈15–35×; B ≈5–20× the standard, depending on route

WVU/ICCT (2014), pp. ii–iii (cold start), 62–65; p. 64 (single runs). VW plea (2017), Exh. 2, ¶ 52 (Exh. 2-21). Ratios are our calculations from published means.

An anomaly still needs testing

Measurement	Check calibration and method agreement
Operating conditions	Check route, temperature, load and regeneration
Inference	Separate a measured gap from its cause

WVU/ICCT (2014), §3.3.3.3 p. 50 and Table 3.13 p. 55; Table 3.15 p. 57; pp. 62–64; Table 4.3 p. 65; p. iv.

What VW did to the test

The “switch” read steering position, speed, engine run time and barometric pressure

Official test recognised

“Dyno calibration”

Full emissions control:
compliant laboratory results

Normal driving

“Road calibration”

SCR / lean-NOx-trap control
weakened on the road

Summary of EPA NOV pp. 3–4 and plea ¶¶ 34, 49–50; a simplification, not the source code

EPA NOV (2015), pp. 3–4; VW plea (2017), Exh. 2, ¶ 34 (Exh. 2 pp. 12–13), ¶¶ 49–51 (Exh. 2 pp. 19–20).

Where the line was crossed

LINE CROSSED

2006

Design decision

Engine could not meet NOx rules and still sell; software built to detect the test

¶¶ 33–36

2009–15

Sales

About 585,000 U.S. vehicles sold with defeat software

¶ 72

2013

Refinement

Steering-angle recognition approved to spot the test

¶¶ 49–50

2014

Independent data

WVU road tests expose the gap; regulators ask why

¶¶ 52–53; EPA p. 4

2014–15

Concealment

False explanations; recall with only limited benefit

¶¶ 54–58; EPA p. 4

Sept 2015

Admission

Defeat device admitted on 3 Sept; EPA notice on 18 Sept

¶¶ 63–64

The line: engineering the product so the official measurement would not represent normal use, then hiding it

VW plea (2017), Exh. 2, ¶¶ 33–36 (Exh. 2 pp. 12–14), 49–58 (pp. 19–23), 59, 63–64 (pp. 23–25), 72 (p. 27). EPA NOV (2015), pp. 1, 4.

What VW wanted, and who paid

What VW wanted	U.S. diesel sales: pass the NOx standard on paper and still attract buyers; later, keep approval for 2016 models
Volkswagen AG	Guilty plea to three felonies: $2.8 billion criminal fine + $1.5 billion civil penalties; 3 years' probation with an independent monitor
Owners and the public	≈585,000 U.S. vehicles; owner settlements and a NOx remediation trust credited at $11 billion; extra NOx exposure not quantified in our sources

VW plea (2017): offences p. 1; monitor p. 12; civil penalties, $11B credit and $2.8B fine p. 13; probation p. 14. Exh. 2 ¶¶ 30–33 (Exh. 2 pp. 10–12), 54–55, 59, 61, 72 (p. 27).

Why this crossed into manipulation

Selective presentation

Choosing favourable observations while leaving other observations out

Documented manipulation

Changing behaviour when the test is recognised and concealing that change from decision-makers

Analytical interpretation grounded in VW plea (2017), Exh. 2 ¶¶ 33–35, 54–55, 63; EPA NOV (2015), p. 4.

What a skeptical analyst should check

1	**Meaning**	Unit, averaging method and denominator
2	**Coverage**	Routes, missing tests and repeat counts
3	**Independence**	Instrument checks and tests outside the standard cycle
4	**Causation**	Alternative explanations, software evidence and admissions

Analytical recommendations based on WVU/ICCT (2014), pp. iv, 55, 57, 59, 64–65, and VW plea (2017), Exh. 2 ¶¶ 33–35, 54–55, 63.

Likely audience questions

Did the road tests prove cheating?

They established the anomaly. Later software evidence and admissions established intent.

Why three vehicles?

Enough to identify cases worth investigating, insufficient to estimate fleet-wide prevalence.

Why use the standard on a road chart?

It provides a clear reference. The routes are not certification tests, and ratios must name their denominator.

Why do sources say 35×, ~40× or 10–40×?

Different scopes: route means (WVU), single runs (plea ¶ 52), drive cycles across the fleet (EPA p. 4).

WVU/ICCT (2014), p. iv, pp. 59, 64–65; EPA NOV (2015), pp. 3–4; VW plea (2017), Exh. 2, ¶ 34 (Exh. 2 p. 13), ¶ 52 (p. 21), ¶ 63.

Primary sources

2014 research

WVU / ICCT report

15 May 2014
Design: pp. 1, 9–10, 50, 55, 57
Standard: Table 4.1 p. 59
Results: p. 64; Table 4.3 p. 65

2015 regulator findings

EPA Notice of Violation

18 September 2015
Switch and calibrations: pp. 3–4
Affected models: p. 5

2017 admissions

Volkswagen plea agreement

11 January 2017, Exhibit 2
¶¶ 30, 33–36, 49–64, 72
Penalties: pp. 12–14

Complete source details and links are also in the accompanying claim register.